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Integrated toposequence-based assessment for sustainable agriculture in the vulnerable region of Bankura district, West Bengal: a GIS approach for food security and policy pathways

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Abstract

Toposequence analysis is essential for optimizing land use, improving soil and water management, enhancing crop productivity, and promoting climate-resilient agriculture. This study employs an Integrated Toposequence Analysis (ITA) framework to evaluate how variations in topography influence soil characteristics, fertility patterns, and cropping systems in Bankura district, located along the fringe of the Chhota Nagpur Plateau in West Bengal. The landscape was classified into upland, midland, and lowland zones using the weighted sum method, based on elevation, relative relief, and slope. Local soil classifications were further associated with the major Indian soil groups of red and yellow soil, lateritic soil, and alluvial soils. Soil chemical properties were analyzed, and a Soil Fertility (SF) map was generated using logistic regression, while Weaver's crop combination method was applied to identify dominant cropping patterns. The results reveal a distinct fertility gradient across toposequence units: lowlands exhibit high fertility, uplands show moderate fertility, and midlands demonstrate relatively low fertility. Irrigation availability is most favorable in lowland agricultural fields, whereas upland regions remain predominantly mono-cropped with paddy. However, improved water management, diversification of cropping patterns, and modification of the crop calendar could significantly enhance yield potential in upland areas. The study highlights the importance of incorporating toposequence-based planning into agricultural decision-making and demonstrates that strategic interventions can strengthen soil productivity and food security. Overall, the findings contribute to sustainable agricultural development and support the achievement of UN Sustainable Development Goal 2 (Zero Hunger).

Keywords Toposequence, Cereal crops, Soil fertility, Sustainable goal, Food security

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Introduction

A toposequence may defined as the spatial and sequential arrangement of topographic positions that maintain the flow connectivity from the higher to lower slopes [27]. This is because these topographic positions are integral parts of the landscape that can be influenced differently by geomorphic and hydrologic processes in the area. When all other factors affecting soil formation remain unchanged, differences in topography lead to the development of distinct types of soils with variations in soil productivity (SP). As a result, topography is one of the key characteristics used to classify soil series [62]. The variation in various soil characteristics is a consequence of different topographic positions, along with the influence of geomorphic activities such as erosion, water movement, the deposition of transported materials, and the movement of water in the area [33]. In this context, the local slope plays a significant role in influencing the soil's chemical, physical, and biological properties, as well as the local drainage capacity of the soil [29]. However, the same soil can become unproductive if reduced conditions persist due to water accumulation [46]. Slope accelerates soil disturbance through erosion and significantly affects soil properties. Differences in topography have an impact on soil processes and development, including soil erosion (SE) and the movement of solutes from higher to lower depths, which have implications for clay migration and layer development in different soils [4]. In soil science, toposequence helps scientists examine how soil properties, such as texture, organic matter content, pH, and nutrient levels, vary with elevation changes [76]. Toposequence reveals variations in soil properties across different elevations or topographic positions. Integrated Toposequence Analysis (ITA) is a method used in environmental and soil sciences to study the variation of soil properties, landforms, vegetation, and hydrology along a toposequence [31]. It examines the relationships between physiographic position, soil properties, and land use along a sequence of topography [61]. ITA has an application for assessing the suitability of land for different uses, developing sustainable land management strategies, and supporting rural development initiatives [52]. ITA is crucial for selecting crops and adjusting soil management practices [68]. Crop varieties should be selected based on specific soil type (ST), and fertilizers or soil amendments should be applied precisely according to soil characteristics [67].

SF can vary significantly across a landscape, and studying SF in the context of a toposequence can provide valuable links to nutrient availability [37]. Variations in nutrient content (e.g., nitrogen, phosphorus, potassium) exist in soils at different elevations or topographic positions [44]. Soils at different elevations can have different nutrient levels due to leaching, erosion, and organic matter accumulation. Soil pH can vary across a landscape, affecting nutrient availability to plants. Different elevations or topographic positions can have soils with varying pH levels and organic matter accumulation [39]. Soils in low lying areas may accumulate more organic matter due to organic material settling, while higher-elevation soils may have less organic matter [59]. Soil microorganisms play a crucial role in nutrient cycling and SF. Microbial populations can differ based on topographic factors, with varying effects on nutrient availability [22]. Some crops may thrive in nutrient-rich soils at lower elevations, while others may be more suitable for nutrient-poor soils at higher elevations [74]. Fertilizer application rates and types can be adjusted to meet the specific nutrient needs of different areas within the field. Understanding these variations allows for more precise and effective soil management practices, improving crop productivity and sustainable agricultural practices [63].

Food security in Indian agriculture remains an ongoing challenge. Still, it is a top priority for the government, and efforts are being made to ensure a stable and sufficient food supply for the country's growing population [70]. The effectiveness of these measures in achieving food security will depend on their successful implementation and adaptability to changing circumstances [26]. West Bengal is one of the leading agricultural states in India. Agriculture plays a significant role in the state's economy, providing livelihoods to a substantial portion of its population [40]. The agricultural landscape in West Bengal is diverse, with various crops being cultivated throughout the year.

The study on toposequence can provide valuable insights into how crop yields or vegetation types vary at different elevations or locations within a landscape [52]. Topography and soil variations allow farmers and agricultural professionals to make informed decisions about crop practices. These insights can lead to more efficient resource use, improved crop yields, and sustainable farming practices [71]. Toposequence data can be integrated into precision agriculture practices, including variable-rate seeding, fertilization, and pesticide application [51]. The toposequence varies the hydrology and soil fertility that control the rainfed rice cultivation in the east Indian plateau [17]. Toposequence study uses quantitative pedogenesis to analyze soil-landform relationships in different physiographic regions, particularly in India's Chhota Nagpur plateau [55]. Studying toposequence can significantly support the United Nations SDGs by contributing to sustainable land use, agriculture, and environmental management. Integrating this knowledge into policies and practices can help advance progress toward

the SDGs and promote a more sustainable and resilient future.

Different researchers have done several studies, such as topography-wise soil formation, vegetation pattern, LULC change, SE and sediment transport, etc. [11, 25, 41, 56, 64]. Integrated toposequence analysis (ITA), which integrates soil maps, soil fertility parameters, and crop zonation to support balanced fertilization policies across different landscape positions, has increasingly incorporated Logistic Regression (LR) models to quantify the influence of topographic and edaphic variables on soil suitability and crop response along toposequences [9, 19, 31]. In soil and agriculture science, ITA study has not been significantly done in agriculture planning and sustainability. Other than different aspects such as soil type and fertility, crop combination, and crop productivity have been analyzed with the ITA, which is not significantly discussed before. So, the objective of the study is to identify the different toposequence zones based on the ITA and soil and toposequence zone-wise planning for sustainable agriculture that supports the United Nations SDGs. In this paper, we analyze a set of scenarios of future agricultural development in the study area and assess potential trade-offs between food production to "prevent hunger (SDG 2), Clean water and sanitation (SDG 6), and Sustainable cities and communities (SDG 11)". The study hypothesizes that toposequence, or topographic sequence, is related to soil types and soil fertility that impacted the agriculture system in the study area. This study is very significant in understanding how pedological ecosystems are organized and react to changes in topography, and it is vital for land management, environmental protection, and scientific research.

Materials and methodology

Study area

Bankura is a district, located in West Bengal, India (Fig. 1a), located between 22°38′ and 23°38′ North and 86°36′ to 87°46′ East and covers a total area of 6882 km^2. It is known for its rich cultural heritage, historical significance, and natural beauty [15]. Bankura is in the western part of West Bengal and shares its borders with Purulia, Paschim Bardhaman, Purba Bardhaman, Hooghly, and Jhargram districts in the west, north, east, and southern directions. Undulating terrain, forests, and numerous hills characterize the district. Bankura district is characterized by a varied terrain, with parts of it being part of the Chhota Nagpur Plateau. The district has undulating hills, plateaus, and valleys, making it a region of great topographical diversity [24]. Several rivers and water bodies crisscross Bankura. The Dwarakeswar River, Shali River, and Kangsabati River are some of the major rivers in the district. In addition to its hilly terrain, Bankura has fertile valleys and plains, particularly along the banks of rivers like the Dwarakeswar, Kangsabati, and Shali [8]. These plains are suitable for agriculture and are used for cultivating various crops. Three types of soils have been found in the study area: red, lateritic, and alluvial. Red soil is classified into many parts, i.e., red grave soil, red sandy soil, and yellow soil [16]. Bankura district has a low to average yield rate for all crops. Some crops have an average yield rate equal to West Bengal. The block-wise yield rate showed that Chhatna, Saltora, Mejia, and Gangajalghati of Bankura Sadar subdivision, Indpur, Hirbandh, and Taldangra of Khatra subdivision, and Sonamukhi, and Joypur of Bishnupur subdivision have low yield rate comparative to the others. Due to the topographical and pedological diversification as well as low agriculture productivity, the Bankura has been selected as a study of toposequence. The alluvial soil is classified in to older alluvial soil, younger alluvial soil and sandy soil. Agriculture is the primary occupation of the people in Bankura. The district is known for cultivating various crops, including rice, pulses, oilseeds, and vegetables.

Database

Several types of data were used to fulfil the objective of the study. SRTM-DEM have been used to create the elevation, relative relief and slope maps of the study area. The NBSS&LUP database and soil health card scheme were used to analyse soil and fertility status respectively. The, soil map was prepared and published by National Atlas of Soil Survey and Land use Planning, Nagpur in co-operation with department of Agriculture, Government of West Bengal, Calcutta at the scale of 1 cm to 5 km. The Indian Government initiated the Soil Health Card Scheme in February 2015, with the objective of distributing soil health cards to farmers following comprehensive soil testing and offering crop-specific fertilizer guidance to them. The soil health card data were available for the farming land plots in the mouzas or villages of Bankura district. In a mouza, five to 24 plots or locations were analyzed by the government of India for the soil health scheme. A total of 3521 mouzas were considered out of 3828 mouzas in the Bankura district. A total of 307 mouzas were excluded due to the locations of forest and barren land. One analysed soil sample per mouza was considered for the study. Thus, 3521 analysed soil chemical property datasets along with their respective locations were obtained from the Soil Health Card database for the Bankura district.

Representative samples were also collected at a soil depth of 0–15 cm from the same location of farmers' filed. A total of 25 soil samples were collected from the field from May 2022 to June 2022. About 400 gm per soil sample was taken into a zipper pouch, labelled as

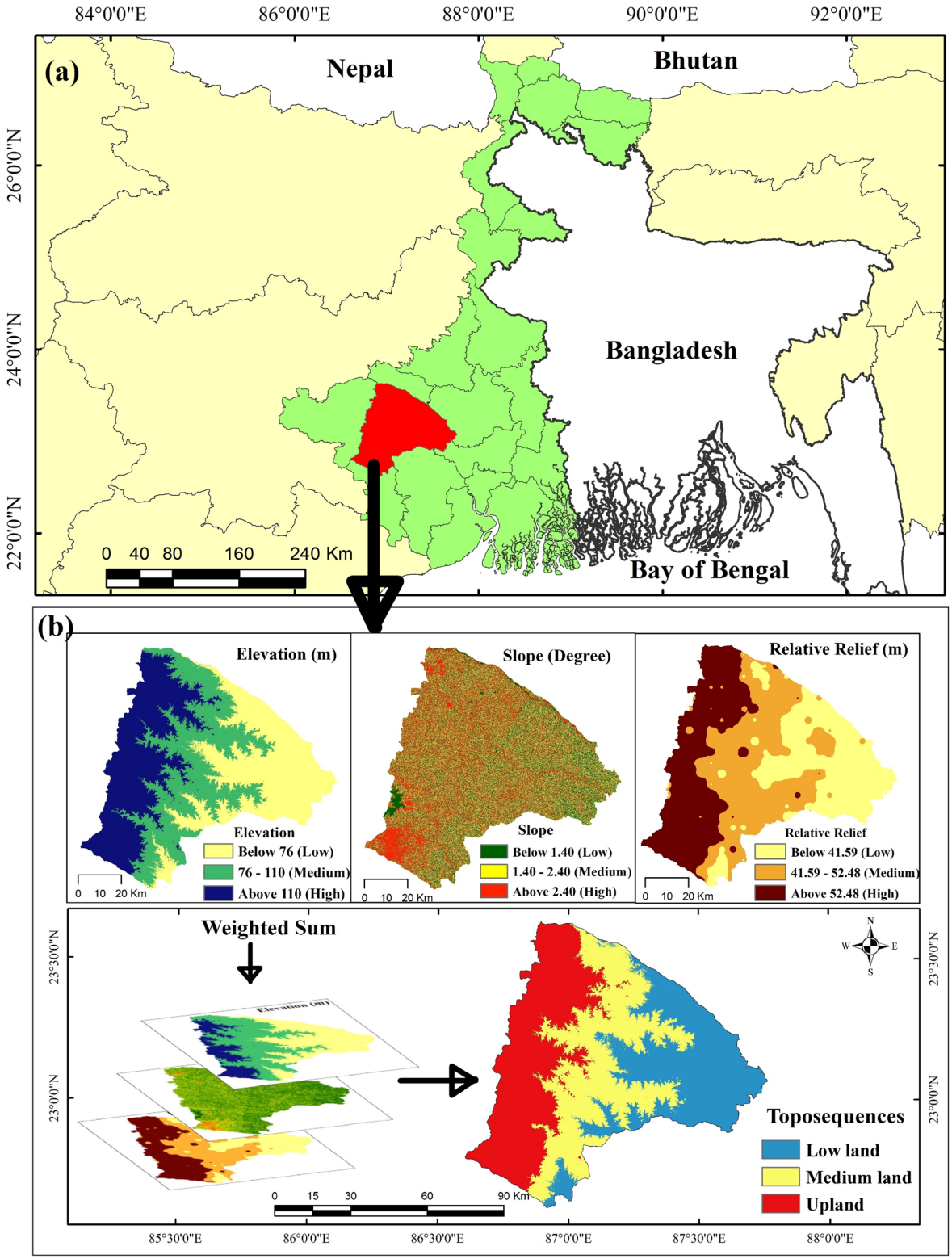


Fig. 1 (a) Location of the study area; (b) Classification of toposequence: Raster layers of elevation, slope, and relative relief have all been added together by applying the weighted sum of the GIS environment

per grid number. The soil samples were then dried in the shade, powdered using a wooden pestle, sieved, and stored in plastic bags for subsequent lab testing. Laboratory-based analysed soil chemical properties were compared with the soil health card database. No significant variation was observed between the chemical properties analysed under the Soil Health Card Scheme and those analysed in the laboratory, and low variability (standard deviation) was observed for the 25 samples. The interpolation tool (Inverse distance weighted method) of geographic information system (GIS) was applied using Soil health card scheme's chemical parameters to determine the spatial distribution individual soil chemical properties maps. Bankura District Statistical Hand Book, 2022 was collected from the Department of Planning, Statistics and Programme Monitoring in West Bengal. UN SDGs and target reports were collected from the UN-published website [65]. The Soil health cards were published in 2015, and recent data were not found, which was identified as a data limitation for the study. Landsat-8 data were used for LULC mapping with a resolution of 30 m for the Bankura district. These data were sourced from "USGS Earth Explorer (https://earthexplorer.usgs.gov/)". The data were collected during the post-monsoon period (30 September 2022). The images were corrected by radiometric and geometric correction. Using the ERDS IMAGINE 9.0 and Sentinel Application Platform (SNAP) 9.0, the images were processed through a series of pre-processing steps, including geometric and radiometric adjustments. The random forest (RF) supervised image classification method was used for LULC classification.

Framework of the present investigation

The detailed methodology was divided into the following steps per study objectives (Fig. 2).

I. The study area was classified into three toposequence zones: high land, midland, and lowland.
II. Three principal STs were extracted from the database of National Atlas of Soil Survey and Land use Planning, NBSS&LUP.
III. The SF zones were delineated with the help of micro and macro elements with help of LR methods.
IV. The Weaver crop combination (CC) method was applied to know the cultivated major crops in three toposequence zones.
V. Mitigation strategies and plans were proposed based on the pathways of the Sustainable Development Goals (SDGs).

From the methodological steps, it was clear that the first three methods aligned with the classification of topography, soil, and soil fertility using various techniques, which were discussed below. The fourth

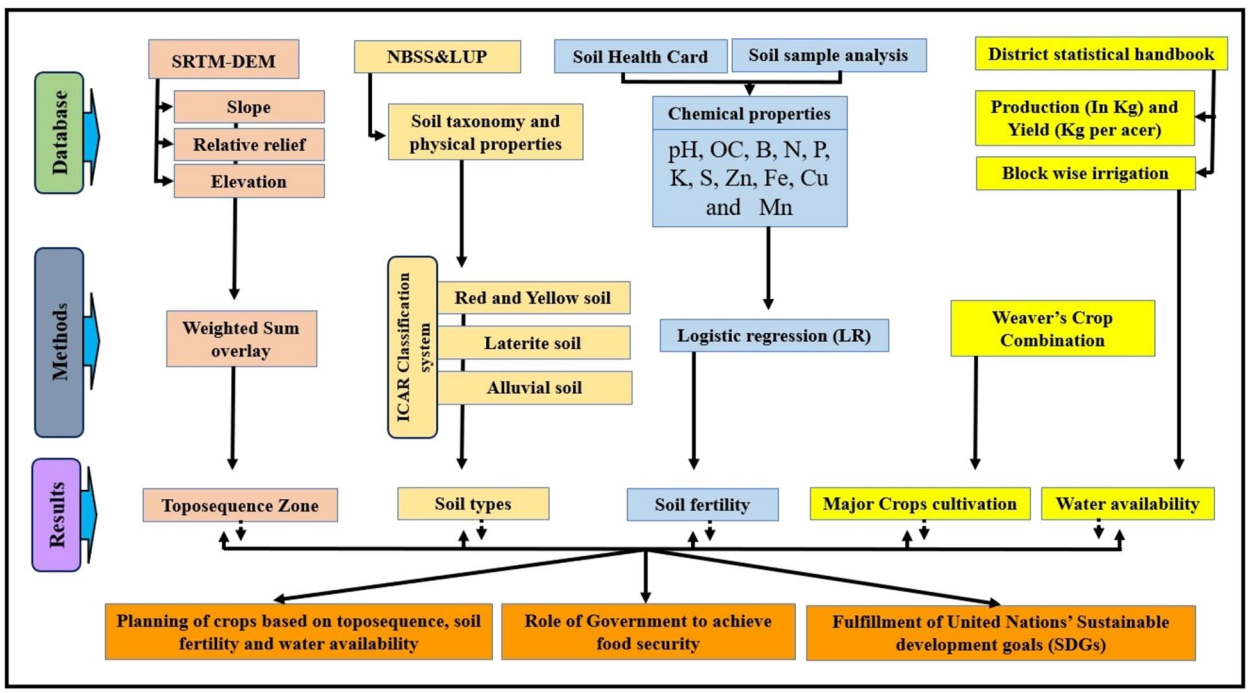


Fig. 2 Methodological flowchart of the study

methodological step addressed present scenarios of crop cultivation or combination. Using this method, toposequence zone-wise planning was carried out for sustainable agriculture that will support the United Nations SDGs.

Logistic regression (LR) model

LR is a statistical model used for binary classification problems where the dependent variable is categorical with two possible outcomes, usually labelled as 0 and 1 [14, 73]. It's a type of regression analysis well-suited for predicting the probability of an event or outcome occurring based on one or more predictor variables. LR was widely used in various fields, including "medicine, finance, marketing, and machine learning" [57], p. [75]. The LR model was estimated the probability that the dependent variable belongs to a particular class (usually 1) as a function of one or more independent predictor variables [58]. The logistic function (also known as the sigmoid function) is used to transform the linear combination of predictors into a value between 0 and 1:

$$P(Y=1)=\frac{1}{1+e^{-(\beta_0+\beta_1X_1+\beta_2X_2+\cdots+\beta_nX_n)}} \quad (1)$$

where: $P(Y=1)$ is the probability of the dependent variable Y being 1. $\beta_0, \beta_{1,\cdots}, \beta_n$ are the model parameters (coefficients) to be estimated. $X_1, X_{2,\cdots}, X_n$ are the predictor variables.

To estimate the model parameters $\beta_0, \beta_{1,\cdots}, \beta_n$ LR uses a maximum likelihood estimation method [54]. Given a set of predictor values, the LR model calculated the probability of the event occurring, and you can then choose a threshold (e.g., 0.5) was chosen to classify the outcome as either 0 or 1 [7]. LR assumes that the relationship between the predictor variables and the log odds of the event was approximately linear [75]. It also assumed that the errors were independent and follow a logistic distribution. LR was considered a robust and interpretable model for binary classification tasks, and it can be extended to handle multiclass classification problems (multinomial LR) as well [18]. The twelve chemical properties and yield (Kg/ha.) of *Aman* paddy were used as independent and dependent variables needed for the LR method, which depicted the SF status of the study area. So, LR was applied for soil fertility mapping using the nominal data of dependent variables that is soil productivity or fertility.

Crop combination

CC, also known as intercropping or companion planting, is an agricultural practice in which two or more different crops are grown together in the same field or garden [48]. This approach is used for various reasons, including maximizing yield, reducing pest and disease pressures, enhancing SF, and conserving space and resources. CC analyses the total percentage acreage area occupied by different crops in a given region in an agricultural year. In agriculture geography, John C. [72] was the first to use statistical techniques to establish the CC analysis in the United States [72]. For the determination of the minimum deviation, the standard deviation method was used:

$$SD=\sqrt{\left(\sum d^2/n\right)} \quad (2)$$

where: d is the difference between the crop percentages in a given county (areal unit) and the appropriate percentage. n is the number of crops in a given combination.

JC Weaver calculated the difference between the hypothetical area (Theoretical) and the actual area of each crop in the combination, and the difference was then statistically represented with the method of standard deviations [36]. The minimum standard deviation value of the respective CC was suggested for that area. The Crop Combination (CC) method was considered valuable for agricultural planning and land use management, but its implementation presents several challenges and limitations. Crop combinations were influenced by climatic variations, which could change unpredictably due to climate change. Additionally, soil fertility differences within a region might not always align with predicted crop combinations. Farmers frequently adjust cropping patterns in response to market demand, government policies, or technological advancements, making it difficult for the method to capture dynamic changes over time. Moreover, the CC method did not always account for external economic factors that influence crop patterns.

Result

Classification of toposequence

A sequence of soils in which distinctive soil characteristics relate to topographic situation. Toposequence may be classified in various ways, including many factors, i.e., elevation-based, slope gradient-based, aspect-based, hydrology-based, LULC-based, etc. [42]. This study classifies the toposequence into three zones, upland, midland, and lowland (Fig. 1b). The GIS based weighted sum method has been applied to sum the elevation, slope and relative relief theme. Weighted sum is a geographical analytic tool that combines several input rasters by multiplying their values by a predetermined weight and then adding them together. This approach effectively prioritizes specific criteria depending on their specified weights, resulting in a final output raster that reflects a comprehensive analysis of the many aspects involved. The weighted sum of elevation, slope and relative relief has been classified into three zones. The breakpoints

to classify the zones quantile interval, as well as field data, have been used. The quantile interval has a benefit in data classification techniques since it divides into groups with an equal number of characteristics, implying that each class will have nearly the same amount of data points. Regardless of the actual value range inside that class, it's especially beneficial for emphasizing the relative ordering of data values rather than the absolute discrepancies between them. The western part is mainly an upland area, and the eastern part is mainly a lowland area between the upland and lowland; there is midland in the study area. The extent of upland, midland and lowland is 2155, 2360 and 2366 sq.km, and the percentage is more or near 30 percent. The study area is known for its diverse landscape and terrain, characterized by a mix of plains, plateaus, and low hills. The western and northern parts of the upland toposequence of Bankura District feature undulating plateaus and uplands. These plateaus are often elevated above the surrounding plains and are characterized by slightly hilly terrain. The eastern part of the lowland toposequence of Bankura District consists of fertile alluvial plains. These plains are well-suited for agriculture and are primarily used for rice cultivation, making agriculture a significant economic activity in this region. Now the middle toposequence is dissected plateau and sometimes plain by the three major rivers of the study area, Damodar, Dwarkeswar and Kangshabati. Some parts of Bankura may feature exposed rock formations, especially In the Upland and midland toposequence. The upland part of the district is covered in dry deciduous forests. These forests are home to various flora and fauna, including 'Sal (*Shorea robusta*)' and 'Teak (*Tectona grandis*)' trees, which are crucial for biodiversity conservation.

Soil forming in toposequence

Bankura district is known for its undulating topography, and variations in ST and properties at different elevations characterize soil formation in this region. Bankura district is mainly found three types of geological division: Chhota Nagpur Genessis complex (Proterozoic era), unclassified metamorphic (Archaean–Palaeoproterozoic era) and older alluvium (Pleistocene) [49]. These rocks include various types of gneisses, schists, and quartzites. The metamorphic rock complex in the district's western part is dominant with the red and yellow soil [60]. The alluvium deposit is mainly found in the middle to the eastern part of the district. laterite soil has occurred in the middle portion of the older alluvium and the east part of the district, where newer alluvium is mainly dominated, and the alluvial soil has been developed. These rock formations can vary in composition and often influence the soil characteristics of the region. The plateaus in this district, which is a metamorphic complex of the western part, have a variety of ST, including red and lateritic soils, due to the differentiation toposequence. In this study, the soil map has been developed based on the soil series of NBSS&LUP with the help of taxonomic characteristics. Similarly, world reference-based (WRB) has been classified as per FAO soil classification (Fig. 4a). Red and yellow soil, laterite soil and alluvial soil, as per the Indian Council of Agricultural Research (ICAR), are the main soil groups [47]. The alluvial soil is mainly formed in lowland topography in the study area characterized by deep to very deep soil depth, very poor to soil drained, loamy and clay soil texture and level to nearly level slope in low laying plain. The red and yellow soil is mainly formed in the highlands in the west and is characterized by shallow to very deep soil depth, moderately well to imperfectly soil drained, fine to gravelly loamy soil surface, and gentle slope to undulating plain or upland on the narrow slope. The laterite soil is mainly dominated in the middle part of the study area, characterized by shallow to deep soil depth, excessively to imperfectly soil drained, coarse to gravelly loamy soil surface and gentle to undulating slope on dissected plateaus or subdued ridges. Like in many other regions, alluvial soils in Bankura district are well-suited for a variety of crops, i.e., rice, pulse, oilseed, vegetables and jute farming, due to their fertility and good water-retaining capacity. Red soils are less suitable for rice cultivation due to their lower water-retention capacity and suitable for a variety of crops, including millets (like sorghum and pearl millet), oilseeds (groundnut and soybean), pulses (like chickpeas and pigeon peas), and cotton [50]. These soils are also used for horticultural crops like fruits and vegetables [23]. Laterite soils are not as fertile as alluvial soils but can still support a range of crops with proper soil management and amendments in low soil water content. Paddy, oilseed, pulses and horticulture farming can be done in the water availability area of laterite soil [28]. The studies show that soil development in the study area was influenced mainly through relief. In the study area, the soil properties were related to topographic aspects and vegetation communities [5].

Soil fertility and toposequence

Toposequence and SF studies are essential for understanding the relationships between soil properties, landscape position, and land use in Bankura District, West Bengal. Researchers or agricultural experts typically select specific study sites within the study area, considering the diverse landscape and ST [38]. To know the status of chemical properties of soil, sites are chosen to represent various topographic positions, such as uplands, midland, and lowlands. From the available nutrients of

the soil, the SF status has been depicted using the LR method. The spatial map of eleven chemical properties as well as soil nutrients and yield (Kg/ha.) of *Aman* paddy has been needed for the SF status of the study area. The Soil pH is acidic, and the optimal level of soil Nitrogen and Manganese) has been found, but there is a minimal amount of organic carbon content in the soil (Fig. 3). The other soil chemical elements, such as Phosphorus, potassium, Boron, Zinc, Sulphur, Iron, and copper have minimal contain in the soil. Nutrient deficiency in soil, such as phosphorus, potassium, zinc, and iron, can reduce crop yields and impede plant development. These shortages can be caused by a variety of causes, including poor drainage, acidic pH, and inadequate watering. An

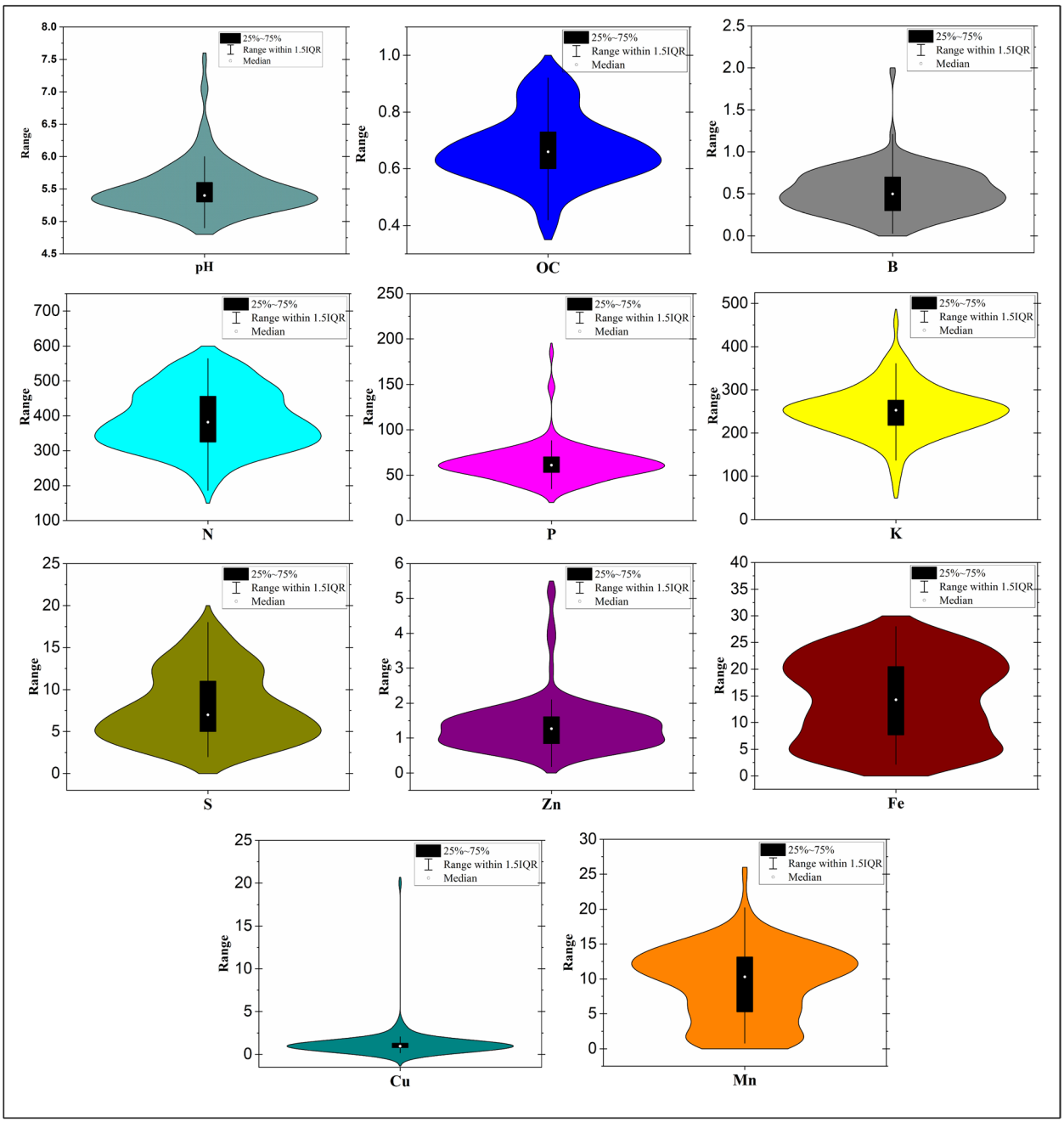


Fig.3 Violin and box plot of; pH, Organic carbon (OC), Boron (B), Nitrogen (N), Phosphorus (P), Potassium (K), Sulphur (S), Zinc (Zn), Iron (Fe), Copper (Cu) and Manganese (Mn) (The part above the red line in the violin graph is the adequate limit of that nutrient. Below the red line indicates insufficient soil nutrients. The standard limit is adopted from ICAR)

overall SF map has been drawn for a holistic relationship between soil nutrient status and toposequence (Fig. 4b). The SF has been classified into three fertility zones. The soil fertility map has also been validated by known random points applied receiver operating characteristic (ROC) methods. The true positive rate and false positive rate of the ROC curve have been used in this regard, and the validation result showed 82.51% accuracy. The classification of soil fertility into three zones based on toposequence, highlighting the low fertility of laterite soils and the improved potential of red and yellow soils with adequate water. It emphasizes optimal nutrient requirements for crops, particularly paddy, and underscores the importance of balanced fertilization and localized soil analysis for improving soil health and productivity. The lowland toposequence is mainly associated with the high fertility of the soil. The high land toposequence zone is related to medium fertility, and midland toposequence is related to low fertility zones. The midland toposequence zone is mainly related to laterite soil, which is unsuitable for any type of agriculture. The red and yellow soil is much better than laterite soil if water is available. Typically, soil should have 30–60 pounds of nitrogen, 15–30 pounds of Phosphorus and 30–60 pounds of Potassium per acre per acre for most crops [10]. Paddy requires higher nitrogen levels, often 60–120 pounds per acre [6]. Phosphorus (P) and Potassium (K) are essential but should be balanced with nitrogen.

Cultivated crop with the toposequence

Farmers traditionally select crops for their land based on a combination of factors, including local knowledge, experience, and observations. Some common traditional methods that farmers use to select crops for their land are soil type, climate, water availability, market demand, risk tolerance etc. Paddy is the main crop of the studied region, and farmers often practice paddy (rice) cultivation. Other than wheat, mustard, lentils and sesame are cultivated all over the district. The toposequence-wise major crops of the block are shown in Table 1. Saltora, Chhatna, Hirbandh, Khatra, Ranibandh, Indpur, Gangajalghati, and Bankura-I belong to upland toposequence, and the crops cultivated in those blocks are *Aman* paddy, wheat, mustard, sesame and lentils. Mejia, Sarenga, Raipur, Bankura-II, Onda and Taldangra belong to midland and the *Boro* paddy and the said upland' crops are the major crops of midland. "Indus, Kotulpur, Joypur, Patrasayer, Bishnupur, Sonamukhi, Simlapal and Borjora" belong to lowland toposequence zone and Aus paddy, potato and jute are included with the major crops of midland. The Weaver CC result showed that the upland and midland were only dominated by *Aman* paddy (Fig. 5a). The block of lowland, Simlapal and Bishnupur, have the fifth number of CCs in a calendar year. Kotulpur and Patrasayer have the fourth number of CCs, Joypur has three, and Sonamukhi and Induas have two number of CCs in a calendar year. So, it is clear that lowland is better for traditional crop practices.

Productivity of the crop with the references of toposequence

The analysis of crop productivity (CP), i.e., yield (Kg per hectare) according to the topography, is important to understand the toposequence-wise crop suitability (Fig. 6 and Table 2). To know this, the season-wise cultivated crops have been selected. The five-year average of yield has been taken for all the crops. All over the district, *Aman* paddy is being cultivated in the kharif season (Jul to Nov). It is that the maximum, minimum and average yield of *Aman* is increasing from upland to lowland. The crops of the Rabi season (Dec to Mar) are mustard, potato, lentil wheat and *Boro* paddy. The highest production has come from the lowland, and the lowest comes from the midland, and the upland showed better productivity than the midland. Aus paddy and sesame are the main cultivated crops of the Zaid season, where the Aus paddy has good productivity in midland and upland than the upland. Sesame (Sesamum indicum) is an important oilseed crop grown for its seeds, which are rich in oil and protein. The production result showed that it has a better productivity (1010 kg/ha.) rate than the middle (843 kg/ha.) and upland (804 kg/ha.). Understanding the toposequence in an agricultural area allows farmers to make informed decisions about which crops to plant at different elevations on the slope. It also helps in planning irrigation and drainage systems to optimize CP. Additionally, microclimatic variations along the toposequence can influence the choice of crop varieties and planting dates to maximize yields and minimize risks. The local crop calendar and planting seasons can determine which crops can be grown at specific times of the year.

Area of the cultivated crop with references to toposequence

Agricultural lands that are located on diverse types of terrain and landforms can have a significant impact on the agricultural practices and crops grown in those areas. Topography includes factors like elevation, slope, aspect, and soil characteristics, and it plays a crucial role in determining the suitability of land for specific types of cultivation. The study area has diversified land, i.e., upland, midland and lowland. The recent LULC map showed that there are 4485 sq.km area of cropland, which is 65.17 per cent of the total area (Fig. 7). The area of cropland in the upland, midland and lowland toposequence zone is 1584, 1230, and 1671 sq. km. In the case of upland, the percentage of cultivation land from the total cropland in the kharif season is 48.84%, but in the

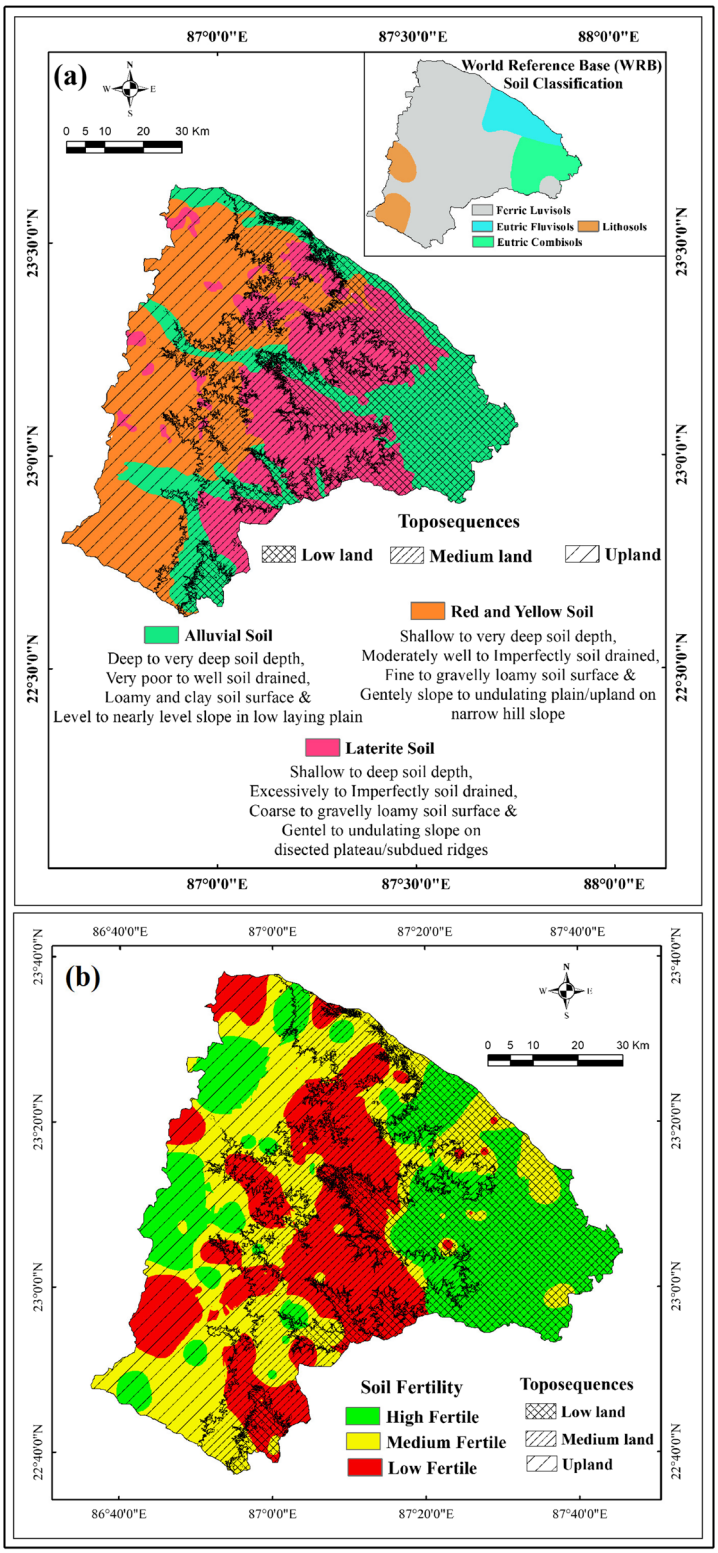


Fig. 4 (a) Soil classification of the study area as per ICAR soil taxonomy; (b) Soil fertility status of the study area: Logistic regression method and eleven soil chemical properties (pH, Organic carbon (OC), Boron (B), Nitrogen (N), Phosphorus (P), Potassium (K), Sulphur (S), Zinc (Zn), Iron (Fe), Copper (Cu) and Manganese (Mn)

Table 1 Present crop practices of farmers with the toposequence; upland and medium land were only dominated by Aman paddy and low land characterized by diversified crops cultivation

Toposequence class	Name of blocks	Cultivated crops
Upland	Saltora, Chhatna, Hirbandh, Khatra, Ranibandh, Indpur, Gangajalghati, and Bankura-I	Aman, Wheat, Mustard, Sesame and Lentils
Medium land	Mejia, Sarenga, Raipur, Bankura-II, Onda and Taldangra	Aman, Wheat, Mustard, Potato, Sesame, Lentils and Boro paddy
Low land	Indus, Kotulpur, Joypur, Patrasayer, Bishnupur, Sonamukhi, Simlapal and Borjora	Aman, Boro, Aus paddy, Wheat, Lentils, Mustard, Sesame, Potato and Jute

case of Rabi and Zaid the percentage is minimal, i.e., 2.91 and 0.237 per cent (Table 3). The percentage of cultivation land in midland from the total cropland in the Kharif season is 62.27%, but in the case of Rabi and Zaid, the rate is minimal, i.e., 13.27 and 6.53%. The percentage of cultivation land in lowland from the total cropland in the Kharif season is 70.77%, but in the case of Rabi and Zaid, the rate is minimal, i.e., 37.38 and 17.49%.

Irrigated area with toposequence

Irrigation can play a significant role in toposequence studies, which involve investigating soil and landscape variations along a slope or topographic gradient. Irrigation can be used to manipulate soil moisture conditions and study how these variations impact different aspects of the ecosystem, such as plant growth, soil properties, and hydrology [69]. The study area is not much facelifted by irrigation in agriculture. Tank, river lifting, open dug well, and shallow tube well irrigation is the main mode of irrigation in the study area. But there are changes in the mode of irrigation toposequence-wise. Canal and shallow tube well irrigation facilities have been applied in the middle and lowland toposequence. Tank and open dug well is the only irrigation mode in the high hilly region. In the highland areas around the district's western boundary and southwestern section (Midland), where hard formation is common, aquifer potential is limited. Ground water abstraction systems in this district consist of dug wells. However, when summer approaches, these drilled wells and small tanks dry up, making further growth impossible in order to manage the drinking water situation. The block-wise percentage of irrigation land showed an unpleasant result for upland and midland (Figs. 5b and 8). The block-wise percentage of irrigation land varies from 6 to 25 per cent in the upland, 25–52 per cent in midland and 52 to 90 per cent in the upland. Most of these parts of the upland area don't have an irrigation facility and are full of grits. Sustainable water management for agricultural development requires addressing challenges like water scarcity and drought by focusing on efficient irrigation techniques, groundwater recharge, and community-based water harvesting to improve crop yields and livelihoods in the upland of Bankura district. This includes implementing rainwater harvesting techniques, such as rooftop rainwater harvesting and small check dams, to collect and store rainwater for irrigation. Additionally, groundwater recharge can be enhanced through measures like constructing recharge wells and percolation tanks.

Implementing irrigation in the diverse topographic settings of the study area presents several challenges due to variations in landscape, soil conditions, and climatic factors. The district primarily depends on monsoon rains; however, uneven rainfall distribution often results in water shortages. Additionally, certain areas face limited groundwater availability due to low water table levels, making groundwater extraction challenging. Large parts of Bankura consist of lateritic and sandy soils with poor water retention, increasing irrigation demand. Furthermore, some low-lying regions experience waterlogging and soil erosion, leading to a decline in soil fertility. Crop selection should be closely aligned with the mode of irrigation you plan to use. Different crops have varying water requirements and adaptability to different irrigation methods. Surface irrigation is suitable for crops that withstand wetting of their leaves and need water to be evenly distributed in the root zone. Rice, cotton, sugarcane, and some vegetables can be grown using surface irrigation. Subsurface irrigation is well-suited for crops with shallow root systems that benefit from consistent soil moisture, i.e., root crops (e.g., carrots and beets).

Planning of crop selection

Toposequence studies are essential for crop selection and CP because they provide valuable insights into the spatial and temporal variations in soil, water, and environmental conditions within a landscape [21]. Understanding these variations, such as topographical, ST, SF, SP, CC, CP, and irrigation, is crucial for making informed decisions about which crops to grow and how to manage them efficiently. First of all, three types of soils are formed, i.e., red and yellow soil, laterite soil and alluvial soil in the study area. Red soil is generally rich in iron and has good drainage properties, while yellow soil may have more sand

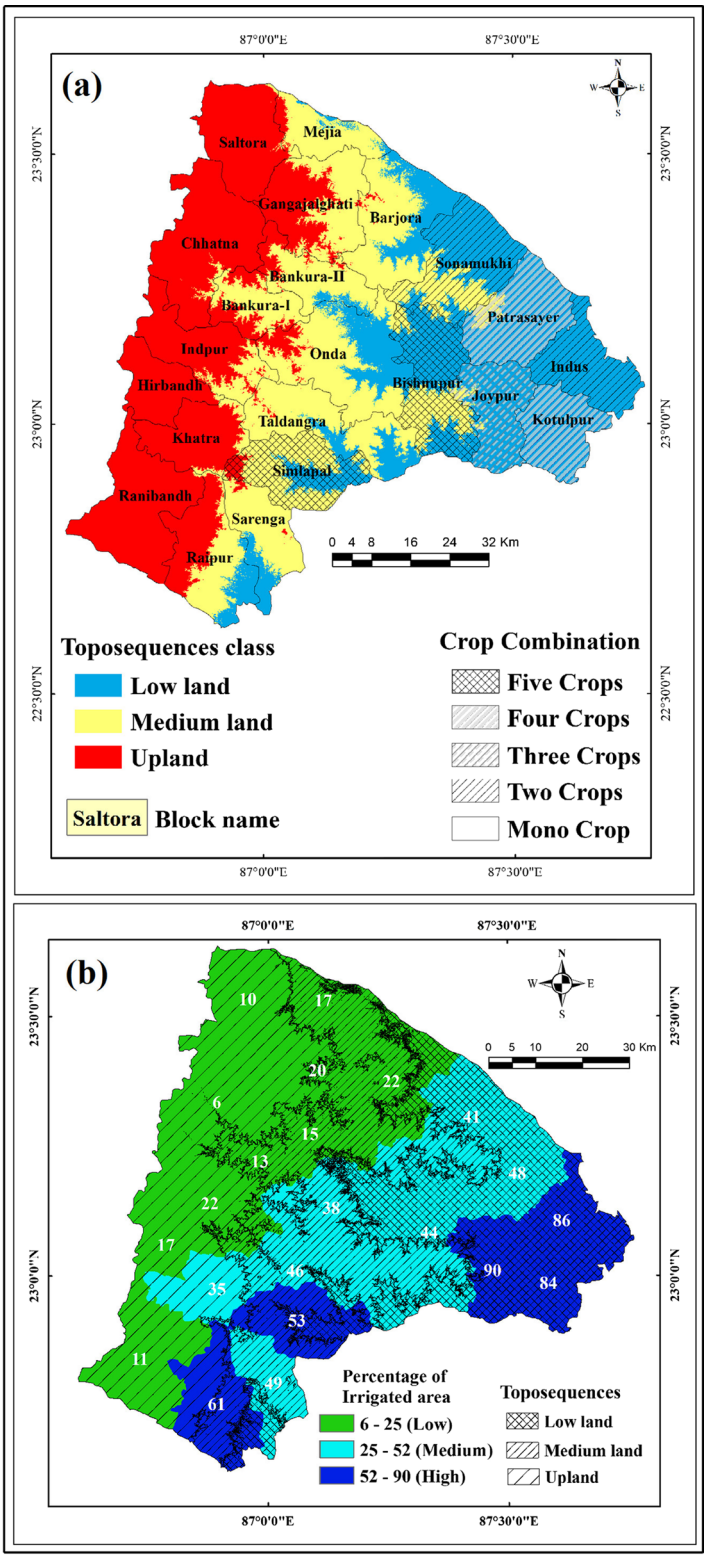


Fig. 5 (a) Toposequence wise crop combination: After John C. Weaver 1954; (b) Percentage of block wise irrigation in the study area

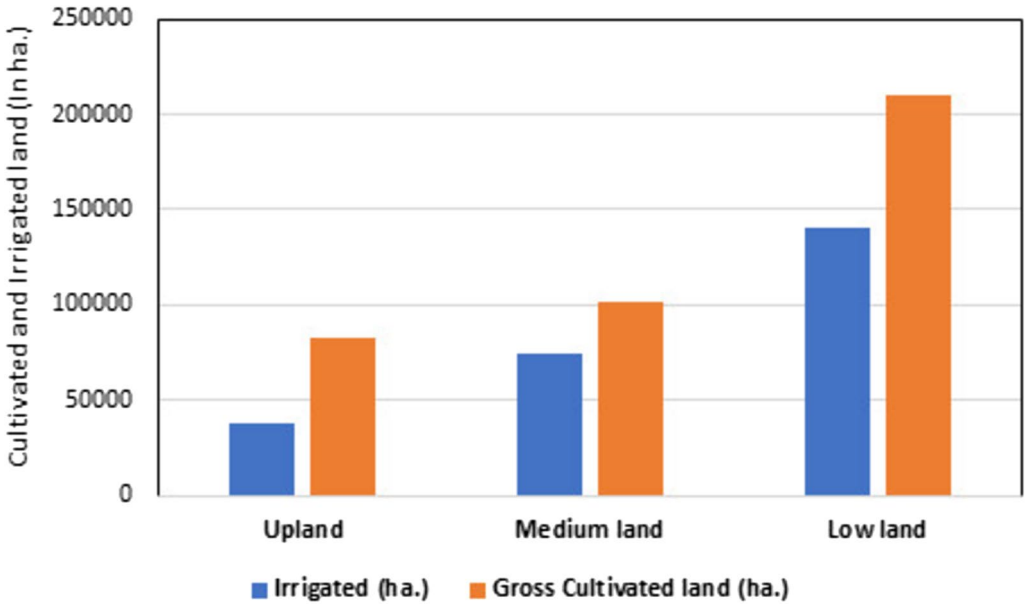


Fig. 6 Toposequence wise irrigated and cultivated land; The cultivated land is increasing with the irrigation in the Bankura district

Table 2 Toposequence wise crop production of seasonal crops

Season	Crop		Yield (Kg per hectare)		
			Upland	Medium land	Low land
Kharif (Jul to Nov)	Aman paddy	Min	1968	2264	2515
		Max	2581	2655	3180
		Avg	2395	2487	2815
Rabi (Dec to Mar)	Mustard	Min	624	472	683
		Max	852	816	1039
		Avg	749	677	799
	Potato	Min	19,932	18,658	21,877
		Max	32,140	35,304	34,354
		Avg	25,414	25,376	28,491
	Lentil	Min	492	511	872
		Max	922	779	1428
		Avg	753	734	1151
	Wheat	Min	1888	2046	2131
		Max	2870	2274	2619
		Avg	2326	2269	2350
	Boro paddy	Min	1989	2793	2618
		Max	3162	3432	3661
		Avg	2604	3040	3000
Zaid (Apr to Jun)	Sesame	Min	877	719	634
		Max	1380	1041	968
		Avg	1010	843	804
	Aus paddy	Min	1481	2309	2341
		Max	2054	2645	2941
		Avg	1768	2504	2679

and silt content. Both red and yellow soils are suitable for rice cultivation where water is available. Sesame and mustard are well-suited for cultivation in red and yellow soils. Other than crops like chickpeas, pigeon peas, and lentils, millet ('Bajra') and finger millet ('Ragi') can thrive in these ST. Laterite soil is common in many parts of the Bankura district; crops like Groundnut (peanut) and soybeans are suitable for laterite soil conditions. These crops can thrive in the well-drained, sandy nature of the soil. Pigeon peas and chickpeas can be suggested in laterite soil with appropriate management of soil acidity. Pearl millet ('Bajra') and finger millet ('Ragi') can tolerate the low fertility of laterite soil and are suitable for cultivation. Cashew nut trees are well-suited to laterite soil conditions and can be a profitable crop for nonarable parts of red and yellow soils. In Bankura district, West Bengal, where alluvial soil is prevalent, farmers have the advantage of being able to grow a variety of crops.

The district receives a significant portion of its annual rainfall during the monsoon season, which has implications for the crops that can be successfully cultivated. The water holding capacity is low in the upland and midland, which is mainly associated with red-yellow and laterite soil, low to medium fertility, and minimal irrigation facilities. Paddy like *Aus* and *Aman* are suitable for rainfed conditions in upland and midland toposequene, but due to uncertainty of monsoon and irrigation, paddy production has not given up to the average. *Aman* rice is primarily cultivated using monsoonal rainfall across the Bankura district. However, due to the uncertainty of rainfall, some rice fields face water shortages during

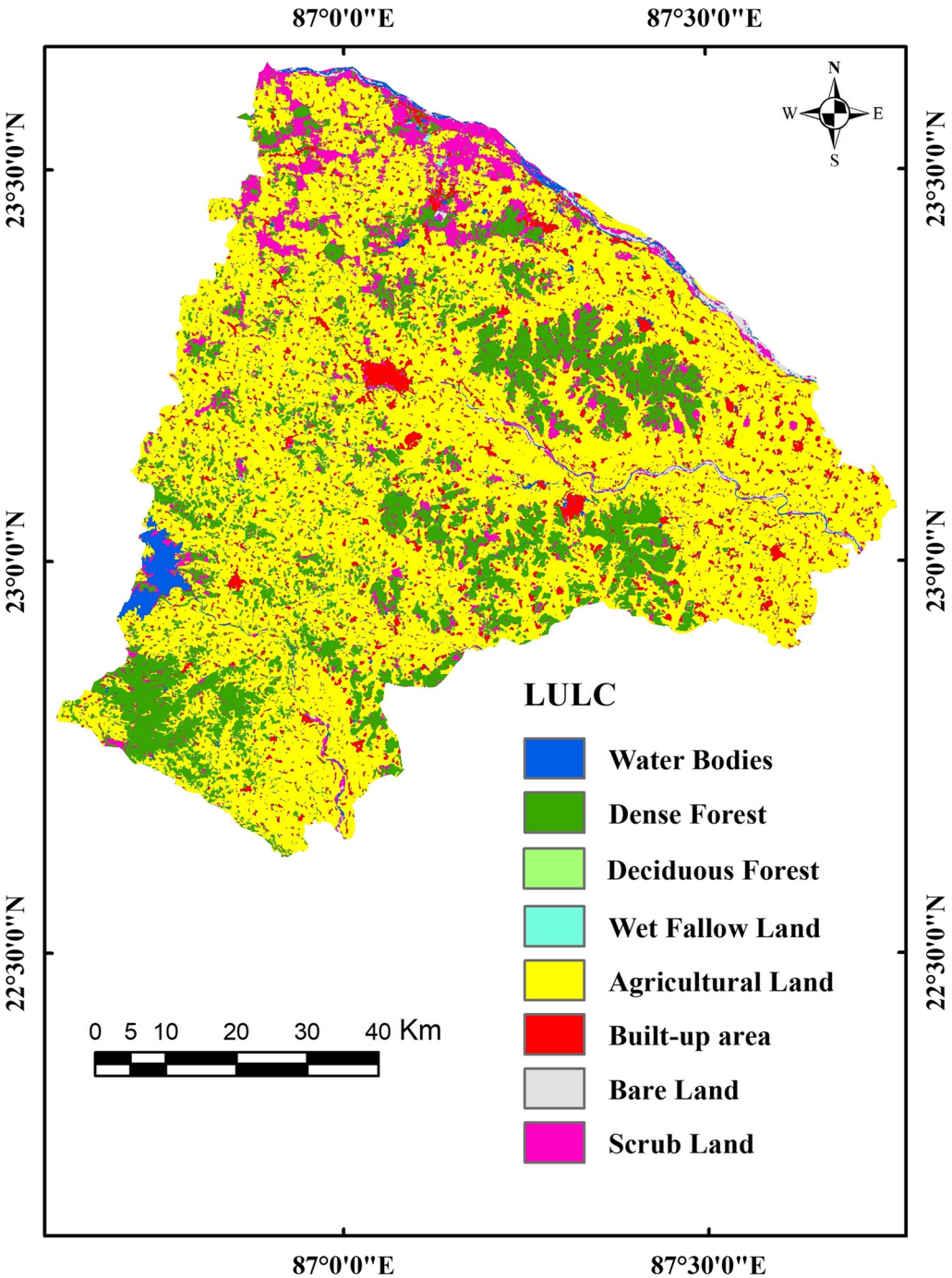


Fig. 7 Toposequence wise irrigated and cultivated land; The cultivated land is increasing with the irrigation in the Bankura district

transplanting and flowering periods. In contrast, certain rice fields with irrigation support achieve higher *Aman* rice productivity. These irrigated fields are mostly located in lowland areas with well-developed irrigation facilities. On the other hand, *Aman* rice fields in upland and midland areas lack adequate irrigation infrastructure. Additionally, the district experiences severe meteorological drought conditions every three to five years [12].

Boro rice, on the other hand, is primarily cultivated using irrigation in the lowland areas of the study region. However, this rice cultivation is often disrupted by pre-monsoon cyclonic storm surges from the Bay of Bengal

Table 3 Area and percentage of cultivated area in different toposequence

Season	Crops	Upland			Medium land			Low land		
		Area (Hectare)	Total area (Hectare)	Percentage of area*	Area (Hectare)	Total area (Hectare)	Percentage of area[a]	Area (Hectare)	Total area (Hectare)	Percentage of area[a]
Kharif (Jul to Nov)	Aman paddy	77,163	77,163	48.84	76,595	76,595	62.27	118,253	118,253	70.77
Rabi (Nov to Mar)	Mustard	1498.58	4601.83	2.91	3443	16,486.75	13.40	5815.75	62,466.41	37.38
	Potato	879.5			4909.5			23,818.25		
	Lentil	28.5			52.167			79		
	Wheat	217.75			1305.5			1688		
	Boro paddy	1977.5			6776.58			31,065.41		
Zaid (Apr to Jun)	Aus paddy	89	376	0.237	2727	8034	6.53	15,910.25	29,240.5	17.49
	Sesame	287			5307			13,330.25		

[a] Percentage is based on total cropland in respective land (Upland = 158,000, Medium land = 123,000 and Low land = 167,100 hectare

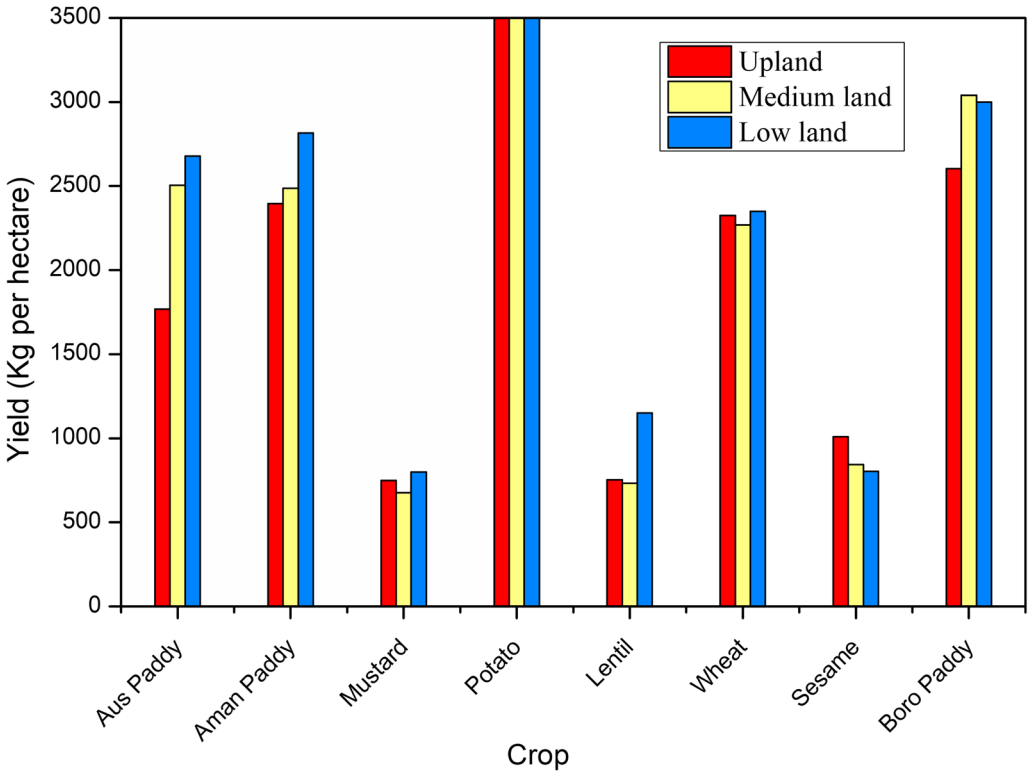


Fig. 8 Toposequence-wise average yield of main crops; Most of the crops showing high yield in low land and comparative low yield in upland except sesame (Source: Bankura District Statistical Hand Book, 2022)

during the harvesting period [13]. Except for paddy cultivation, the fallow land can be utilized for planting the crops of Rabi and Zaid seasons; for example, crops like chickpeas, pigeon peas, and green gram (moong dal) are commonly grown in rainfed areas; Groundnut and sesame are oilseed crops that can thrive in rainfed conditions in the monsoon season. They are drought-resistant and can be important sources of cooking oil and protein. Rainfed vegetables like okra (Ladyfinger), bottle gourd, and bitter gourd can also be grown successfully during the monsoon season.

Discussion

A toposequence study, also known as a topographic sequence study, is a type of research that focuses on the relationship between topography (the physical features and elevation of the land) and various environmental factors [21]. It is an essential component of landscape analysis and has applications in fields such as ecology, hydrology, and land management [34]. Landform analysis reveals the relationships between morphological, physical, and chemical soil properties across different units, showing how location influences these properties and reflects terrain conditions [2]. As the need for timely and relevant soil information to support sustainable soil resource management grows, digital soil mapping (DSM) has gained popularity across various ecosystems including arid, semi-arid, and humid regions, as well as rangelands and forests due to its ability to address the limitations of traditional soil mapping methods [1]. This study aims to understand how topographic variations influence soil, environmental and agricultural patterns. In this study, different topographic positions, such as upland, midland and lowland, exhibit significant variations in soil properties, including texture, organic matter content, pH, and nutrient availability. Understanding these variations is crucial for optimizing soil management practices, such as fertilization and irrigation [30]. There is a growing demand for agricultural productivity to meet the increasing needs of a rapidly expanding global population [35]. This challenge is particularly pronounced in countries like India, where the population growth rate is high. Conducting standard soil surveys is a systematic approach to gathering information about soils, including their origin, extent, and characteristics. It is vital for assessing their potential and limitations for specific purposes [66]. In managing soil resources within different topographic positions (toposequences), knowledge about the physical and chemical attributes and classifications of soils is an indispensable [20]. Adegbite

et al. [3] found that three soil pedons collected from various toposequences had significant variations in physical, chemical, and morphological features, and their classes were established based on these characteristics. Gobin et al. [32] [32] established correlations between physiographic location, soil qualities, vegetation, and land use to highlight the physical and human impacts on the existing land condition along the catenae. Their findings further demonstrate that there is little opportunity for additional agricultural intensification on the upland interfluve regions with current farming system methods, as well as a growing demand to utilize fluvial wetlands.

The productivity of crops in eastern India is greatly constrained by impoverished SF caused mainly by poor soil nutrient management strategies. The presence of soil nutrients exerts a notable influence on agriculture and the environment, especially concerning aspects such as SF, SQ, and rice cultivation. Within agricultural ecosystems, "pH, Boron (B), Nitrogen (N), Phosphorus (P), Potassium (K), Sulphur (S), Zinc (Zn), Iron (Fe), Copper (Cu), Manganese (Mn) and Organic Carbon (OC)" are the major indicators and determinants of SQ and fertility as they are strongly linked to plant growth and productivity. Paddy is being cultivated across the three toposequences in the study area, and the mentioned chemical properties have been associated with the yield of paddy by the LR method, which is a new approach for spatial mapping of SF. Variations in essential nutrients such as nitrogen (N), phosphorus (P), potassium (K), and micronutrients significantly impact crop productivity in the region. The predominance of lateritic soils, which are low in organic matter, results in poor nitrogen retention, affecting crops like *Aman* and *Boro* rice. In upland areas, highly weathered soils often experience phosphorus fixation, leading to reduced productivity of legumes, pulses, and oilseeds. Continuous cropping without adequate nutrient replenishment depletes potassium levels, thereby lowering crop resilience to climatic stress. Additionally, zinc and iron deficiencies are prevalent in paddy fields, particularly in irrigated *Boro* rice cultivation. Frequent meteorological droughts (occurring every 3–5 years) further exacerbate the issue by reducing soil moisture, thereby limiting nutrient uptake by crops [12]. The study of Dessalegn et al. [21] also indicates that upland areas are less fertile than lowland areas because of erosion and the eluviation-illuviation process. Irrigation is crucial in ensuring food security in India, a country with a large and growing population. It allows farmers to grow multiple crops per year, increasing agricultural productivity and enabling the cultivation of a wider variety of crops. That's why the lowland area is more productive and diversified in terms of crops in the study area, Bankura district. While irrigation is essential, there are challenges related to over-extraction of groundwater, water quality issues, and environmental impacts. Thus, reliance solely on irrigation expansion in upland and midland areas is not sustainable, but modifying the crop calendar and adopting suitable alternative crops is one step towards the "UN SDGs goal". CC is the method to know the crop diversification in an area. The present study area is dominated by a single crop, namely *Aman* paddy. The toposequence-wise productivity analysis shows that upland and midland are mainly dominated by paddy. Other crops, such as Aus paddy, potato, sesame, mustard and wheat, are also cultivated in the lowland area in the district. The toposequence-wise result indicates that upland and midland areas, mainly red, yellow and laterite, low percentage of irrigated land, single crop and low productivity.

Food security by the United Nation (UN)

The "United Nations (UN)" plays a crucial role in ensuring food security around the world through various agencies, programs, and initiatives [43]. Food security is defined as the state where all people, at all times, have physical, social, and economic access to sufficient, safe, and nutritious food to meet their dietary needs and food preferences for an active and healthy life [45]. The FAO is a specialized agency of the UN that leads global efforts to eradicate hunger and improve food security. It provides technical assistance and expertise to countries in developing sustainable agriculture, fisheries, and forestry practices. The FAO conducts research, data collection, and policy advocacy to address global food security challenges. The "UN's Sustainable Development Goal 2 (SDG 2)" is dedicated to ending hunger, achieving food security, improving nutrition, and promoting sustainable agriculture ("UN SDGs," 2016). The "2030 Agenda and its 17 SDGs" provide a framework for global action on food security and related issues. This study supports "SDGs 2 (Zero hunger), SDG 6 (Clean water sanitation) and SDG 11 (Sustainable cities and communities)" (Fig. 9). Target 2.4 is part of the broader effort to address food security and agriculture in a way that not only ensures access to sufficient and nutritious food for all but also promotes environmental sustainability and resilience in the face of climate change and other challenges. It emphasizes the need for sustainable and resilient agricultural practices and the protection of ecosystems and natural resources. The study directly supports improvements in agricultural productivity and yield using toposequence-wise analysis that fulfills the target 2.4. In ITA-based agricultural planning, farmers can adopt climate-resilient and sustainable practices to cope with climate change, extreme weather, and disasters. The study findings suggest promoting drought-resistant crops, agroforestry, and organic farming to enhance soil fertility in upland areas. Additionally,

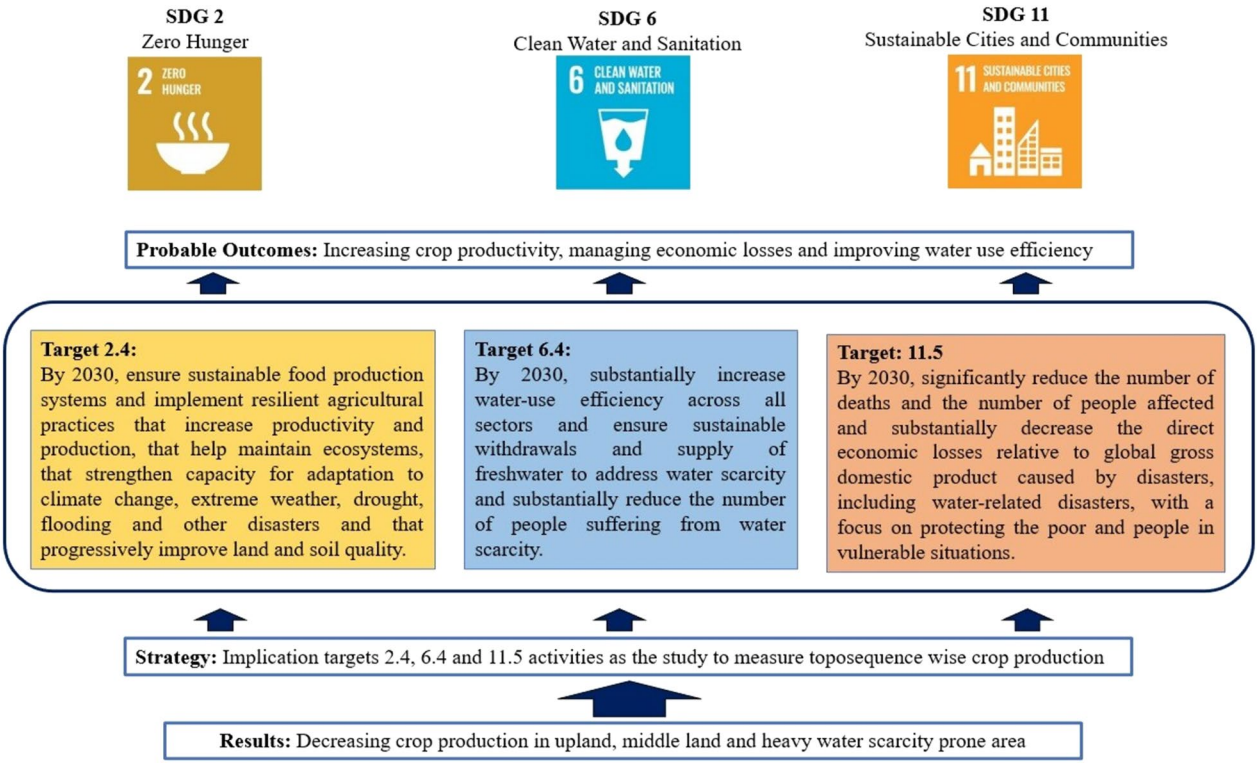


Fig. 9 Priority chart of SDGs' target concerning the status of increasing crop production in upland and middle land

crop rotation, intercropping, and balanced fertilizer use are encouraged to maintain productivity in midland areas. In lowland areas, the focus should be on cultivating water-tolerant crops, implementing efficient drainage systems, and adopting controlled irrigation to mitigate waterlogging. Target 6.4 focuses on improving the efficient use of water resources across various sectors, such as agriculture, industry, and households, to address water scarcity. This study contributes to improved surface and groundwater management in water-scarce upland areas. Target 11.4 focuses on reducing the impact of disasters, including natural disasters and other types of crises, on cities and communities. It seeks to mitigate the loss of life, reduce the number of people affected by disasters, and minimize the economic losses associated with these events. Indirectly, the findings contribute to reducing economic losses in the study area by enhancing resilience to environmental and climatic risks.

Governments employ a range of policies to bolster the food and agriculture sectors. These policies encompass trade and market interventions, such as border measures and price controls, which create either price incentives or disincentives. Additionally, governments provide fiscal subsidies to producers and consumers and extend general services support, as Anderson (2022) noted. These policies have far-reaching consequences, impacting all participants in the food industry, and influencing the accessibility and affordability of nutritious diets within the larger food landscape. Indian government has implemented several to achieve food security or zero hunger, such as "Integrated Child Development Services (ICDS), Food for Work (FFW), and Public Distribution System (PDS)", etc. [53]. But these programs need huge amounts of food grains. India is self-dependent on food grain production, but it is not enough to increase its population. Such food grain distribution mechanisms are not a fully sustainable long-term solution. Therefore, optimal utilization of potential agricultural land should be prioritized by the government. The effectiveness of repurposing food and agricultural policy support will also be shaped by the political environment, the concerns of various stakeholders, the level of market power concentration, and the governance structures and regulatory frameworks that facilitate the reform process (Fig. 10). There are many potential challenges in implementing the mentioned scheme, such as socio-economic and administrative barriers, these are: slow processing of agricultural schemes and financial assistance, which leads to delays in benefits reaching farmers; leakages in fund allocation and improper execution of programs, which hinder effectiveness; lack of coordination among different government departments and agencies, resulting in fragmented

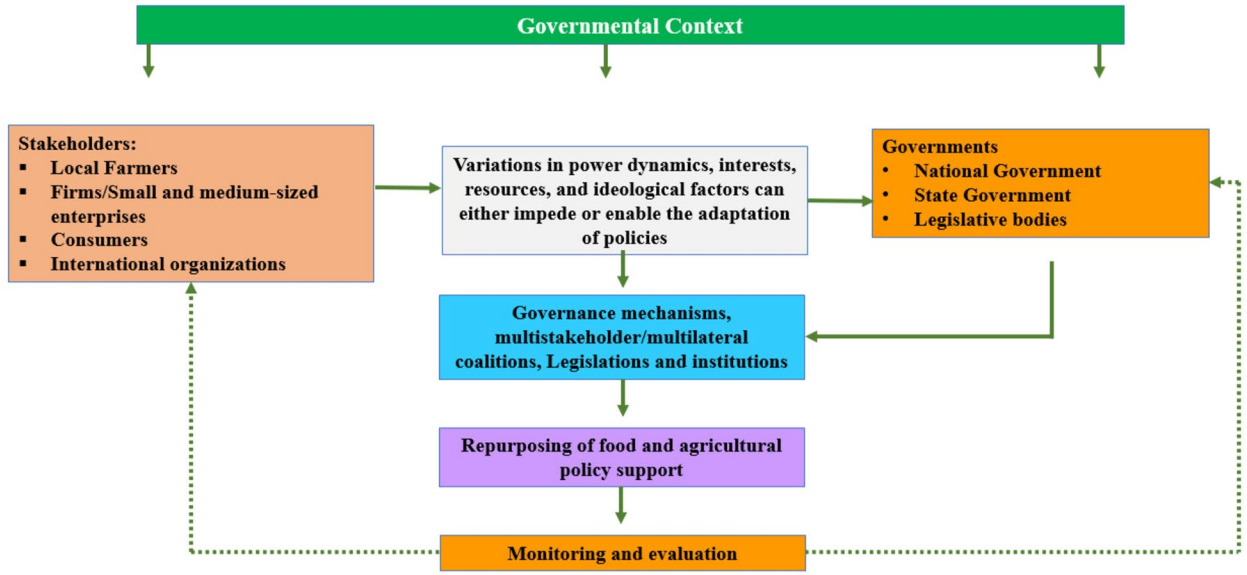


Fig. 10 Role of Government to achieve food security as per FAO

implementation; exploitation by middlemen and limited access to direct markets, which prevent farmers from receiving fair prices; and limited awareness among farmers regarding government subsidies, credit facilities, and modern agricultural techniques. There are also several limitations in implementing the proposed recommendations, such as data limitations or community acceptance. A lack of accurate and up-to-date hydrological and soil data may hinder effective planning and implementation of water management strategies. Local communities may resist adopting new techniques due to limited awareness, strong adherence to traditional practices, or concerns regarding costs. Constructing rainwater harvesting structures, recharge wells, and percolation tanks requires a significant initial investment, which may be unaffordable for small farmers. Additionally, limited technical expertise can result in improper implementation and poor maintenance of water harvesting and recharge structures.

Conclusion

The ITA study has provided valuable insights into the diverse agricultural potential of the landscape under investigation. The toposequence exhibits pronounced spatial variability in soil characteristics, including texture, organic matter content, and nutrient availability across upland, midland and lowland units. This variability directly impacts crop suitability and yield potential for major and alternate crops. Topography plays a crucial role in water distribution and drainage where higher elevations tend to be well-suited for rain-fed agriculture, while lower-lying areas may benefit from controlled irrigation to manage excess moisture. The SF pattern aligns with the topographical division, with lowland areas exhibiting highly fertile soil and diversified crops. Based on our findings, we propose alternative cropping systems and adaptive crop calendar strategies for sustainable agriculture. The alternative crops for the upland areas, characterized by red and yellow soil, include millets, pulses, sesame, and cashew nut, which are drought-resistant, suited to dry conditions, and can enhance soil fertility through nitrogen fixation. The midland, characterized by an undulating terrain, is suitable for maize, vegetables (brinjal, chilli, tomato, ginger, and turmeric), and agroforestry, which can reduce soil erosion and improve land stability. The lowland alluvial plains are well-suited for sugarcane, jute, and aquaculture, which can enhance farmers' income and improve water-use efficiency. These strategies encompass crop selection, soil management practices, water resource utilization, and land-use planning. Effective implementation requires collaboration between local communities, agricultural practitioners, and policymakers. A toposequence-based agricultural plan in Bankura can enhances drought resilience in uplands through water conservation and drought-tolerant crops, while flood-prone lowlands can benefit from flood-tolerant cropping and aquaculture. By integrating crop zoning, agroforestry, soil conservation, and water management, the district can ensure sustainable livelihoods and climate resilience. To successfully implement drought resilience and flood management strategies in Bankura district, collaboration among government agencies, farmers, research institutions, NGOs, and local

communities is essential. The findings of this study promote sustainable and resilient farming practices, as well as the conservation of ecosystems and natural resources. The study significantly contributes to agricultural planning by applying toposequence-wise analysis to improve productivity, yield, and food security. Furthermore, the outcomes of this research can guide future studies in land-use planning, agricultural extension services, and conservation efforts. Recognizing the importance of sustainable farming practices and resilience-building measures is paramount. By integrating the knowledge gained from this toposequence study, we can better address the challenges of climate change and variability.

List of abbreviations

B	Boron
CC	Crop Combination
CP	Crop Productivity
Cu	Copper
FAO	Food And Agriculture Organization
Fe	Iron
FFW	Food for Work
GIS	Geographic Information System
ICAR	Indian Council of Agricultural Research
ICDS	Integrated Child Development Services
ITA	Integrated Toposequence Analysis
K	Potassium
LR	Logistic Regression
LULC	Land Use and Land Cover
Mn	Manganese
N	Nitrogen
NBSS&LUP	National Bureau of Soil Survey and Land Use Planning
OC	Organic Carbon
P	Phosphorus
PDS	Public Distribution System
RF	Random Forest
S	Sulphur
SNAP	Sentinel Application Platform
SDGs	Sustainable Development Goals
SE	Soil Erosion
SF	Soil Fertility
SP	Soil Productivity
SQ	Soil Quality
SRTM-DEM	Shuttle Radar Topography Mission- Digital Elevation Model
ST	Soil Type
UNs	United Nations
WRB	World Reference-Based
Zn	Zinc

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Author contributions
I.C. and S.C.P conceived the idea for the research paper and led the overall direction and planning. I.C. also conducted the literature review, collected data and contributed to the writing of the manuscript. S.C.P. provided critical feedback, edited the manuscript and contributed to the interpretation and presentation of the findings. All authors have read and approved the final version of the manuscript. All authors reviewed the manuscript. "C.B.P., A.I. contributed to review writing whereas the final drafting was done by A.R.M.T.I., E.A. and M.K.I.

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Competing interest
The authors declare no competing interests.

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